

Query Results

Q1-Find all orders where sales are greater than the average sales. (Subquery) ->

Average sales-

average_sales
234.41818199917006

```
1 • SELECT * FROM orders
2 WHERE sales >
3 (
4     SELECT AVG(sales)
5     FROM orders
6 )
```

row_id	order_id	order_date	ship_date	ship_mode	customer_id	product_id	sales	quantity	discount
1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	FUR-BO-10001798	261.96	2	0
2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0
4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	FUR-TA-10000577	957.5775	5	0.45
8	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002275	907.152	6	0.2
11	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	FUR-TA-10001539	1706.184	9	0.2
12	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	TEC-PH-10002033	911.424	4	0.2
14	CA-2016-161389	12-05-2016	12-10-2016	Standard Class	IM-15070	OFF-BI-10003656	407.976	3	0.2
17	CA-2014-105893	11-11-2014	11/18/2014	Standard Class	PK-19075	OFF-ST-10004186	665.88	6	0
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	FUR-TA-10000577	1044.63	3	0
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	FUR-BO-10004834	3083.43	7	0.5
36	CA-2016-117590	12-08-2016	12-10-2016	First Class	GH-14485	TEC-PH-10004977	1097.544	7	0.2
39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	FUR-BO-10002545	532.3992	3	0.32
41	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	TEC-PH-10000486	371.168	4	0.2
55	CA-2016-105816	12-11-2016	12/17/2016	Standard Class	JM-15265	TEC-PH-10002447	1029.95	5	0
58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	FUR-CH-10003968	319.41	5	0.1
68	CA-2014-106376	12-05-2014	12-10-2014	Standard Class	BS-11590	OFF-AR-10002671	1113.024	8	0.2
73	US-2015-134026	4/26/2015	05-02-2015	Standard Class	JE-15745	FUR-CH-10000513	831.936	8	0.2
86	CA-2017-140088	5/28/2017	5/30/2017	Second Class	PO-18865	FUR-CH-10000863	301.96	2	0
104	US-2015-156867	11/13/2015	11/17/2015	Standard Class	LC-16870	TEC-AC-10001552	238.896	6	0.2

Q2- Find the highest sales order for each customer. (Subquery) ->

```

1 • SELECT * FROM orders o
2   WHERE sales =
3   (
4     SELECT MAX(sales)
5     FROM orders
6     WHERE customer_id = o.customer_id
7   );

```

Result Grid										
Filter Rows:										
Export: Wrap Cell Content:										
row_id	order_id	order_date	ship_date	ship_mode	customer_id	product_id	sales	quantity	discount	
2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	FUR-CH-10000454	731.94	3	0	
4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	FUR-TA-10000577	957.5775	5	0.45	
11	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	FUR-TA-10001539	1706.184	9	0.2	
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	FUR-TA-10000577	1044.63	3	0	
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	FUR-BO-10004834	3083.43	7	0.5	
36	CA-2016-117590	12-08-2016	12-10-2016	First Class	GH-14485	TEC-PH-10004977	1097.544	7	0.2	
39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	FUR-BO-10002545	532.3992	3	0.32	
55	CA-2016-105816	12-11-2016	12/17/2016	Standard Class	JM-15265	TEC-PH-10002447	1029.95	5	0	
58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	FUR-CH-10003968	319.41	5	0.1	
68	CA-2014-106376	12-05-2014	12-10-2014	Standard Class	BS-11590	OFF-AR-10002671	1113.024	8	0.2	
110	CA-2015-129476	10/15/2015	10/20/2015	Standard Class	PA-19060	TEC-AC-10000844	339.96	5	0.2	
146	CA-2015-110744	09-07-2015	09-12-2015	Standard Class	HA-14920	OFF-ST-10003656	671.93	7	0	
150	CA-2016-114489	12-05-2016	12-09-2016	Standard Class	JE-16165	FUR-CH-10000454	1951.84	8	0	
166	CA-2014-139892	09-08-2014	09-12-2014	Standard Class	BM-11140	TEC-MA-10000822	8159.952	8	0.4	
174	CA-2014-118962	08-05-2014	08-09-2014	Standard Class	CS-12130	FUR-CH-10003817	340.144	7	0.2	
182	CA-2014-166191	12-05-2014	12-09-2014	Second Class	DK-13150	TEC-AC-10004659	408.744	7	0.2	
190	CA-2015-102281	10-12-2015	10/14/2015	First Class	MP-17470	FUR-BO-10002613	899.136	4	0.2	
219	CA-2016-130162	10/28/2016	11-01-2016	Standard Class	JH-15910	TEC-PH-10002563	302.376	3	0.2	
224	CA-2015-169397	12/24/2015	12/27/2015	First Class	JB-15925	TEC-MA-10001148	479.988	4	0.7	
227	CA-2015-163055	08-09-2015	8/16/2015	Standard Class	DS-13180	FUR-TA-10003748	622.45	5	0	

Q3- Calculate total sales for each customer. (CTE) ->

```
1 • WITH customer_sales AS
2   (
3     SELECT
4       customer_id,
5       SUM(sales) AS total_sales
6     FROM orders
7     GROUP BY customer_id
8   )
9   SELECT * FROM customer_sales;
```

Result Grid			Filter Rows:	Export:
	customer_id	total_sales		
▶	CG-12520	1148.7800000000002		
	DV-13045	1119.4830000000002		
	SO-20335	2602.5750000000001		
	BH-11710	6255.3510000000001		
	AA-10480	1782.872		
	IM-15070	4930.4739999999999		
	HP-14815	1990.314		
	PK-19075	8158.654		
	AG-10270	2565.25799999999994		
	ZD-21925	1493.944		
	KB-16585	2742.75800000000003		
	SF-20065	763.54599999999998		
	EB-13870	2767.219		
	EH-13945	684.171		
	TB-21520	4730.628		
	MA-17560	4280.6209999999999		
	GH-14485	1361.24400000000001		
	SN-20710	3127.95920000000007		
	IC-16930	4481.162		

Q4- Find customers whose total sales are above average. (CTE + Subquery) ->

```
1 • WITH customer_sales AS
2   (
3     SELECT
4       customer_id,
5       SUM(sales) AS total_sales
6     FROM orders
7     GROUP BY customer_id
8   )
9   SELECT * FROM customer_sales
10  WHERE total_sales >
11  (
12    SELECT AVG(total_sales)
13    FROM customer_sales
14  );
```

	customer_id	total_sales
▶	BH-11710	6255.351000000001
	IM-15070	4930.473999999999
	PK-19075	8158.654
	TB-21520	4730.628
	MA-17560	4280.620999999999
	SN-20710	3127.9592000000007
	LC-16930	4481.162
	RA-19885	3832.3139999999994
	ES-14080	4648.516
	KM-16720	4909.472000000001
	KD-16270	8282.358
	JE-15745	6400.454999999999
	SC-20770	4492.656999999999
	CS-12400	3651.8585000000007
	GM-14455	3078.6240000000003
	RB-19465	4556.5660000000002
	GZ-14470	4355.15
	JM-15250	3145.3740000000003
	CL-12565	10880.545999999998
	SC-20725	5226.2099999999998
	AD-10180	6100.999999999999
	LS-16975	2988.674
	LC-16885	7396.7739999999985
	JD-15895	6456.456
	SC-20000	2406.581

Q5- Rank all customers based on total sales. (Window Function) ->

```

1 • WITH customer_sales AS
2   (
3     SELECT
4       customer_id,
5       SUM(sales) AS total_sales
6     FROM orders
7     GROUP BY customer_id
8   )
9   SELECT
10    customer_id,
11    total_sales,
12    RANK() OVER (ORDER BY total_sales DESC) AS customer_rank

```


	customer_id	total_sales	customer_rank
▶	SM-20320	25043.05	1
	TC-20980	19017.847999999998	2
	RB-19360	15117.339	3
	TA-21385	14595.62	4
	AB-10105	14355.610999999997	5
	SC-20095	14142.333999999999	6
	KL-16645	14071.917	7
	HL-15040	12873.297999999999	8
	SE-20110	12209.438000000002	9
	CC-12370	12129.072	10
	TS-21370	11885.871	11
	GT-14710	11820.119999999997	12
	BM-11140	11609.900000000001	13
	SV-20365	11420.645999999997	14
	CJ-12010	11079.742	15
	CL-12565	10880.545999999998	16

Q6-Assign row numbers to each order within a customer. (Window Function + PARTITION BY) ->

```

1 • SELECT
2     customer_id,
3     order_id,
4     sales,
5     ROW_NUMBER() OVER
6     (
7         PARTITION BY customer_id
8         ORDER BY sales DESC
9     ) AS row_num
10    FROM orders;

```

	customer_id	order_id	sales	row_num
▶	AA-10315	CA-2016-103982	3930.072	1
	AA-10315	CA-2014-128055	673.568	2
	AA-10315	CA-2016-103982	431.976	3
	AA-10315	CA-2017-147039	362.94	4
	AA-10315	CA-2014-128055	52.98	5
	AA-10315	CA-2016-103982	41.72	6
	AA-10315	CA-2015-121391	26.96	7
	AA-10315	CA-2014-138100	14.94	8
	AA-10315	CA-2014-138100	14.56	9
	AA-10315	CA-2017-147039	11.54	10
	AA-10315	CA-2016-103982	2.304	11

Q7- Display top 3 customers based on total sales. (Window Function) ->

```
1  WITH customer_sales AS
2  (
3      SELECT
4          customer_id,
5          SUM(sales) AS total_sales
6      FROM orders
7      GROUP BY customer_id
8  )
9  SELECT * FROM
10 (
11     SELECT
12         customer_id,
13         total_sales,
14         RANK() OVER (ORDER BY total_sales DESC) AS customer_rank
15     FROM customer_sales
16 ) ranked_customers
17 WHERE customer_rank <= 3;
```

customer_id	total_sales	customer_rank
SM-20320	25043.05	1
TC-20980	19017.847999999998	2
RB-19360	15117.339	3

QUESTION: Final Combined Query

Write one final query that shows:

- Customer Name
- Total Sales
- Rank (Use **JOIN + CTE + Window Function** together) ->

```

1 • WITH customer_sales AS
2   (
3     SELECT
4       customer_id,
5       SUM(sales) AS total_sales
6     FROM orders
7     GROUP BY customer_id
8   ),
9   customer_details AS
10  (
11    SELECT
12      customer_id,
13      MAX(customer_name) AS customer_name
14    FROM customers
15    GROUP BY customer_id
16  )
17  SELECT
18    cd.customer_name,
19    ROUND(cs.total_sales, 2) AS total_sales,
20    RANK() OVER (ORDER BY cs.total_sales DESC) AS customer_rank
21  FROM customer_sales cs
22  JOIN customer_details cd
23  ON cs.customer_id = cd.customer_id
24  ORDER BY customer_rank;

```

	customer_name	total_sales	customer_rank
►	Sean Miller	25043.05	1
	Tamara Chand	19017.85	2
	Raymond Buch	15117.34	3
	Tom Ashbrook	14595.62	4
	Adrian Barton	14355.61	5
	Sanjit Chand	14142.33	6
	Ken Lonsdale	14071.92	7
	Hunter Lopez	12873.3	8
	Sanjit Engle	12209.44	9
	Christopher Conant	12129.07	10
	Todd Sumrall	11885.87	11

Mini Project: Customer Sales Insights

Q1- Who are the top 5 customers?

	customer_name	total_sales
▶	Sean Miller	25043.05
	Tamara Chand	19017.85
	Raymond Buch	15117.34
	Tom Ashbrook	14595.62
	Adrian Barton	14355.61

```
1 • WITH customer_sales AS
2   (
3     SELECT
4       customer_id,
5       SUM(sales) AS total_sales
6     FROM orders
7     GROUP BY customer_id
8   ),
9   customer_details AS
10  (
11    SELECT
12      customer_id,
13      MAX(customer_name) AS customer_name
14    FROM customers
15    GROUP BY customer_id
16  )
17
18  SELECT
19    cd.customer_name,
20    ROUND(cs.total_sales,2) AS total_sales
21  FROM customer_sales cs
22  JOIN customer_details cd
23  ON cs.customer_id = cd.customer_id
24  ORDER BY total_sales DESC
25  LIMIT 5;
```


Q2- Who are the bottom 5 customers?

	customer_name	total_sales
▶	Thais Sissman	4.83
	Lela Donovan	5.3
	Mitch Gastineau	12.32
	Carl Jackson	16.52
	Roy Skaria	22.33

```
1 • WITH customer_sales AS
2   (
3     SELECT
4       customer_id,
5       SUM(sales) AS total_sales
6     FROM orders
7     GROUP BY customer_id
8   ),
9   customer_details AS
10  (
11    SELECT
12      customer_id,
13      MAX(customer_name) AS customer_name
14    FROM customers
15    GROUP BY customer_id
16  )
17
18  SELECT
19    cd.customer_name,
20    ROUND(cs.total_sales,2) AS total_sales
21  FROM customer_sales cs
22  JOIN customer_details cd
23  ON cs.customer_id = cd.customer_id
24  ORDER BY total_sales ASC
25  LIMIT 5;
```

Q3- Which customers made only one order?

```

1 • WITH customer_details AS
2   (
3     SELECT
4       customer_id,
5       MAX(customer_name) AS customer_name
6     FROM customers
7     GROUP BY customer_id
8   )
9
10  SELECT
11    cd.customer_name,
12    COUNT(o.order_id) AS total_orders
13  FROM orders o
14  JOIN customer_details cd
15  ON o.customer_id = cd.customer_id
16  GROUP BY cd.customer_name
17  HAVING COUNT(o.order_id) = 1;

```

Result Grid			Filter Rows:	Export:	Wrap C
	customer_name	total_orders			
▶	Jocasta Rupert	1			
	Mitch Gastineau	1			
	Anthony O'Donnell	1			
	Lela Donovan	1			
	Carl Jackson	1			
	Ricardo Emerson	1			

Q4- Which customers have above-average sales?

	customer_name	total_sales
▶	Sean Miller	25043.05
	Tamara Chand	19017.85
	Raymond Buch	15117.34
	Tom Ashbrook	14595.62
	Adrian Barton	14355.61
	Sanjit Chand	14142.33
	Ken Lonsdale	14071.92
	Hunter Lopez	12873.3
	Sanjit Engle	12209.44
	Christopher Conant	12129.07
	Todd Sumrall	11885.87
	Greg Tran	11820.12
	Berkv Martin	11609.9

```

1 • WITH customer_sales AS
2 (
3     SELECT
4         customer_id,
5         SUM(sales) AS total_sales
6     FROM orders
7     GROUP BY customer_id
8 ),
9 customer_details AS
10 (
11     SELECT
12         customer_id,
13         MAX(customer_name) AS customer_name
14     FROM customers
15     GROUP BY customer_id
16 )
17
18 SELECT cd.customer_name,
19        ROUND(cs.total_sales,2) AS total_sales
20 FROM customer_sales cs
21 JOIN customer_details cd
22 ON cs.customer_id = cd.customer_id
23 WHERE cs.total_sales >
24 (
25     SELECT AVG(total_sales)
26     FROM customer_sales
27 )
28 ORDER BY total_sales DESC;

```

Q5- What is the highest order value per customer?

	customer_name	highest_order_value
▶	Sean Miller	22638.48
	Tamara Chand	17499.95
	Raymond Buch	13999.96
	Tom Ashbrook	11199.968
	Hunter Lopez	10499.97
	Adrian Barton	9892.74
	Sanjit Chand	9449.95
	Bill Shonely	9099.93
	Sanjit Engle	8749.95
	Christopher Conant	8399.976

```
1 • WITH customer_details AS
2 (
3     SELECT
4         customer_id,
5         MAX(customer_name) AS customer_name
6     FROM customers
7     GROUP BY customer_id
8 )
9
10 SELECT
11     cd.customer_name,
12     MAX(o.sales) AS highest_order_value
13 FROM orders o
14 JOIN customer_details cd
15 ON o.customer_id = cd.customer_id
16 GROUP BY cd.customer_name
17 ORDER BY highest_order_value DESC;
```

Brief Insights-

- Top customers contribute significantly more sales than the rest.
- Several customers have sales above the average, indicating high-value buyers.
- Some customers placed only one order and may need better engagement.
- Customer ranking helps identify the most profitable customers.
- SQL features like CTEs and window functions simplify complex sales analysis